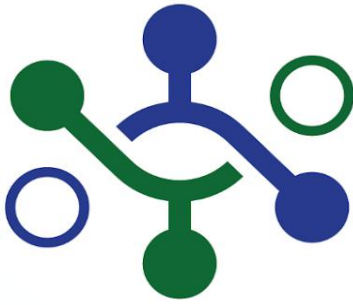




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DARE-RC

DATA AND RESEARCH IN EDUCATION
RESEARCH CONSORTIUM

جراتِ تحقیق ملے

DARE-RC

Call for Applications - Cycle 4

**DARE-RC SCOPING PAPER/EVIDENCE SYNTHESIS TEMPLATE AND
GUIDELINES**



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DARE-RC Scoping Paper/Evidence Synthesis Template and Guidelines

The scoping paper/evidence synthesis proposal should be submitted by the deadline mentioned in the [DARE-RC Call for Applications – Cycle 4 document](#). The scoping paper/evidence synthesis proposal should include five sections, as detailed below.

Note: All documents should be in standard Times New Roman font and size 11 (1.15 spacing).

The scoping paper/evidence synthesis proposal will have five sections:

1. Project summary information
2. Lead organisation and project team
3. Proposal details
4. Main proposal
5. Documents to be attached

Key Information about the Applicant Organisation: Please provide the following information.

Name of Applicant Organisation: Lahore University of Management Sciences	
Type of Applicant Organisation (Public, Private, Research Think Tank, NGO, Others-please specify): Private University	Geographical Scope (where will the research be conducted in Pakistan): 2 Selected Provinces
Address of Applicant Organisation: DHA Phase 5, Lahore, Pakistan	Contact Number of Applicant Organisation: +92 42 3560 8477
Applicant Organisation's website address: www.lums.edu.pk	Name & Email Address of Lead Investigator (s): Dr. Zubair Khalid, Zubair.khalid@lums.edu.pk
Application Number (If this is the first application, please enter "1" and continue the sequence accordingly for multiple applications):	In case of Joint Venture-please specify the name(s) of the other partner entities: None

1. Project Summary Information

1.1. Project Title:

Understanding Climate-Related Educational Disruptions in Pakistan Through the Lens of Data, Inequality, and Policy

1.2. Programme of Work:

Climate resilient education systems - evidence synthesis Total Budget: 10,341,980 PKR

1.3. Project Start and Completion Dates:

April 1, 2026- September 07, 2026

2. Lead Organisation and Project Team

2.1. Lead Organisation:

This contract, if awarded, will be led by Syed Babar Ali School of Science and Engineering (SBASSE), Lahore University of Management Sciences (LUMS), in close collaboration with the Syed Ahsan Ali and Syed Maratib Ali School of Education (SOE) and the Centre for Urban Informatics, Technology, and Policy (CITY) at LUMS.

LUMS is a leading university in Asia offering cutting-edge knowledge and research in various fields. With a faculty strength of over 300 and a student size of 5000+, the University is known for its high-quality undergraduate and graduate programmes, cutting-edge research and development in business, natural and social sciences, law, and most recently, education.

SBASSE brings LUMS' core strength in science, engineering, and computing, and is described by LUMS as the first private research school of science and engineering in Pakistan, with a “no-boundaries” approach that supports cross-disciplinary collaboration and world-class research. SOE is LUMS' school dedicated to strengthening education leadership, policy, and practice in Pakistan. It offers graduate programmes including the MPhil in Education Leadership and Management (ELM) as its flagship programme, alongside other offerings that build education-sector capacity. CITY, based at LUMS, complements this by serving as a hub for urban innovation, integrating data-driven insights, technology, and partnerships to support climate-resilient and sustainable development.

2.2. Other FCDO Contracts:

Lahore University of Management Sciences (LUMS) is currently party to three active contracts with the UK FCDO through the DARE-RC programme:

- Follow-on Funding for Forest fire detection and early warning system: September 2023 – March 2024
- Teacher learning as a mediator of teaching quality at primary school in Khyber Pakhtunkhwa: February 2025 – September 2026
- Scaling Impact: Large-Scale Human Capital Interventions for Low-Income Children in Punjab: May 2025 – September 2026

The total combined value of these three contracts is approximately PKR 96.9 million (approximately £270,000), which falls under £1 million.

2.3. Lead Investigator:

Dr. Zubair Khalid holds a Ph.D. in Electrical Engineering from Australian National University. He is an Associate Professor at the School of Science and Engineering and Co-Director of the Centre for Urban Informatics, Technology & Policy (CITY) at LUMS. His research focuses on AI for climate resilience, spatial data analytics, and climate modeling. He was a recipient of the Gordon Bell Prize (Nobel of Supercomputing) for his work on the development of exascale climate emulators and has secured over USD 4.3 million in research funding in the past five years, including the project (funded by EU) focused on strengthening Pakistan's higher-education resilience and digital learning capacity through policy development, faculty training, and e-learning infrastructure.

2.4. Co-Investigators:

Dr. Jessica Albrent holds an EdD from Johns Hopkins University and is an assistant professor at the Syed Ahsan Ali and Syed Maratib Ali School of Education at Lahore University of Management Sciences (LUMS). Her research agenda focuses on using data and assessments to support improved opportunities to learn for traditionally marginalized communities. She regularly advises the Punjab government on policies and practices to support effective assessments and works with private schools on developing school improvement models. Her past and current research projects include examinations of graduate program outcomes, the influence of teacher-training reforms in classrooms, the development of computational simulations of family-level decision-making on school enrollment, and data use practices in private schools. Dr. Albrent is responsible for the design of the project and the implementation and analysis of the qualitative portion of the project.

Dr. Mohammad Mansoor Khan holds a Ph.D. in Educational Policy and Leadership from the State University of New York (SUNY) at Albany and is an Assistant Professor at the School of Education, LUMS. His work focuses on curriculum development, educational leadership, internationalization, and public policy, with a particular interest in using AI for curriculum development and climate change education. Previously, at SUNY System Administration, he designed and managed state-funded grant programs totaling over \$50 million and supported research policy development for hiring in advanced technology areas, including AI and cybersecurity. He is currently contributing to climate change education training for the Government of Punjab and develops courses in data analysis, educational leadership, and research methods, drawing on both quantitative and qualitative expertise.

3. Project Details (500 words)

3.1. Project Summary (max. 100 words):

Pakistan's education system faces increasing disruption from extreme climate events (floods, heatwaves, and smog) with impacts disproportionately affecting vulnerable populations. This scoping study will systematically map and synthesize existing administrative data, policies, and research evidence to clarify how climate shocks intersect with socioeconomic inequality to affect educational access/outcomes. Over six months, the team will assess data availability and quality, identify evidence gaps, and evaluate feasible analytical approaches for linking climate, education, and SES indicators. Outputs include a comprehensive scoping report, policy brief, and stakeholder roundtable, providing decision-makers with actionable recommendations for building climate-resilient, equitable education systems across Pakistan's provinces.

3.2. Staff Duties (max. 100 words):

The PI, Dr. Zubair Khalid, will lead the project, overseeing identification, assessment, and analysis of administrative datasets and guiding decisions on analytical approaches. Co-PIs Dr. Jessica Albrent and Dr. Mansoor Khan will lead policy and literature reviews, support dataset analysis, and integrate findings with theoretical frameworks. Two part-time graduate research assistants will assist with document mapping, data extraction, and preliminary analysis. All team members will collaborate on synthesis, drafting outputs, and stakeholder engagement. Responsibilities are distributed to ensure efficient progress across all stages while maintaining high-quality, evidence-informed deliverables.

3.3. Policy Engagement Plan (max. 100 words):

Stakeholders from the two provinces identified in Stage 1 (details in Section 4.3), alongside federal education and disaster management authorities, will be engaged throughout to validate findings and identify data gaps. Dissemination will include a focused half-day roundtable, a concise policy brief, and summary notes shared via DARE-RC's platforms. Target audiences are provincial and federal policymakers, education planners, NGOs, and researchers. Outputs will include a scoping report, policy brief, and peer-reviewed article, ensuring evidence informs planning, highlights inequities, and supports equitable education strategies under extreme climate events, while enabling DARE-RC to integrate findings into broader action and learning initiatives.

3.4. Value for Money (max. 100 words):

This six-month scoping study provides strong value for money by delivering a systematic synthesis of climate, education, and SES data, identifying gaps, and mapping patterns of disruption across Pakistan. Resources requested are proportional to the study's scope, supporting document review, dataset assessment, and policy analysis. Outcomes will inform improvements in data systems, reporting, and linked decision-making, providing high-quality, actionable insights for policymakers and education planners. By clarifying analytical possibilities and evidence gaps, the study maximizes impact relative to cost and lays a foundation for future modeling and planning initiatives.

3.5. Ethics Statement and Plan (max. 100 words):

This project involves no primary data collection and poses minimal ethical risk. All data sources will be either publicly available (published policies, reports, academic literature) or anonymized administrative datasets where individual participants are not identifiable. The team will obtain necessary permissions from data custodians and adhere to all data protection regulations. Analytical models developed to generate synthetic data or simulate scenarios will use only aggregated, de-identified inputs to ensure no individual-level information is reconstructed. Stakeholder consultations will be conducted with informed consent. All data will be stored securely, and findings reported in aggregate form to prevent re-identification of individuals or institutions.

4. Main Proposal (1000 words)

4.1. Introduction (max. 350 words):

i. Background and Problem Statement

Pakistan's education system is increasingly affected by extreme climate events, including floods, smog, and heatwaves, yet there is limited understanding of how these events intersect with socioeconomic inequality to influence educational access and outcomes. Globally, over 400 million students have experienced climate-related school closures since 2022, with low- and middle-income countries

disproportionately affected (World Bank, 2024). In South Asia, extreme heat and flooding have shortened school days and slowed learning, particularly impacting lower-income students and increasing risks for girls, including school dropout and early marriage (Ahmed et al., 2019; UNICEF, 2023, 2024).

In Pakistan, access to education is highly stratified by socioeconomic status. For example, 84% of children in the poorest quintile in Punjab lack foundational reading skills compared with 41% in the richest (UNICEF, 2022), yet national-level evidence on how climate disruptions interact with SES is scarce. Existing research and policy largely examine isolated events or provinces, leaving gaps in understanding patterns of vulnerability, the role of government responses, and effective interventions.

This scoping review will synthesize administrative data, published literature, and policy documents to identify patterns of climate-related educational disruption, clarify what evidence exists, and highlight gaps in research and policy. The findings will inform policymakers, education planners, and researchers seeking to improve equitable access and resilience across Pakistan's education system.

ii. Purpose of the scoping paper/evidence synthesis

The evidence synthesis will synthesize existing research, policies, administrative data, and documented practices to clarify what is known about how extreme climate events affect educational access in Pakistan. It will identify consistent patterns, gaps, and feasible intervention pathways to guide further analysis and inform policy decisions.

iii. Policy and Sector Relevance

This work will provide decision-makers with a clear picture of how climate shocks interact with socioeconomic inequality to disrupt education, and what data and interventions can realistically guide more resilient planning. The issue is timely as Pakistan faces increasing extreme temperatures, flooding, and smog, yet lacks integrated evidence to inform response strategies. Likely beneficiaries include provincial and federal education departments, disaster management authorities, school system leaders, donors, and researchers working on climate resilience and equitable access.

4.2. Guiding Research Questions (max. 50 words):

Q1. What climate, education, and SES data exist in Pakistan, and what gaps limit understanding of educational disruption?

Q2. In what ways do extreme climate events influence academic achievement and educational access across different socioeconomic groups in Pakistan?

Q3. What analytical approaches, interventions, and policies show promise for educational access during extreme climate events?

4.3. Proposed Methodology (max. 400 words):

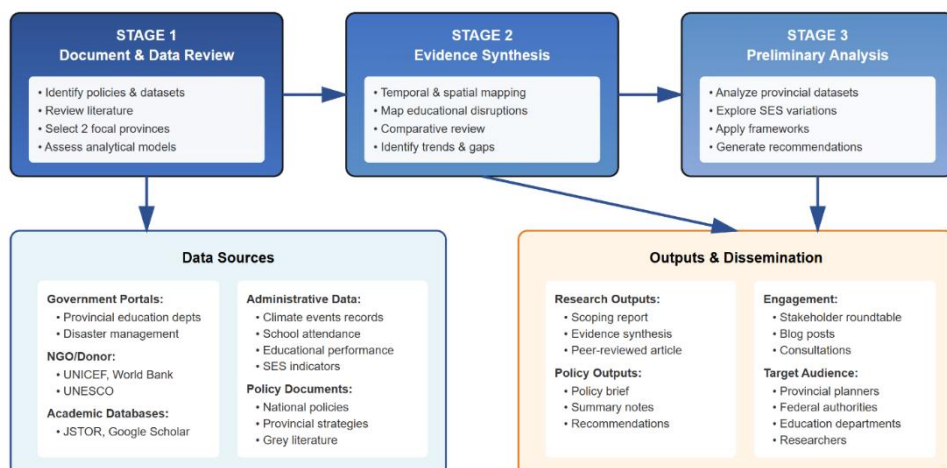
This scoping study will combine document review, data mapping, and analytical synthesis to examine how extreme climate events intersect with socioeconomic status to shape educational outcomes in Pakistan.

Stage 1: Systematic Document and Data Review: The team will identify and review national and provincial policies, administrative datasets on climate events, educational outcomes, and SES, as well as relevant literature. The team will use the 6-stage framework for a scoping paper (Arksey & O'Malley,

2005; Peters et al., 2020), starting with the research questions, identifying relevant studies, selecting the studies, charting the data, synthesizing and reporting the results, and then consulting with key informants. , using the Population, Concept, and Context framework. Populations will include K-12 schools and children and vulnerable communities; the concept of the search will examine intersections between extreme climate events, educational access, and SES-related inequities; and the context will be examined in three stages. First, we will examine broad evidence from the Global South, then move to South Asia, to Pakistan specifically, with a final step in looking at evidence at the provincial level. Scoping studies allow for broad inclusion of a wide range of data, so search strategies will draw on government portals, NGO and donor repositories, and academic databases, while structured templates will guide extraction of key information on data sources, policy measures, and reported outcomes. Information will be organized into themes and summarizing the literature (and gaps in the literature) based on these themes at each level, ranging from the international to the provincial. We will also assess potential analytical models and frameworks for later stages, including approaches suitable for incomplete/heterogeneous datasets, such as descriptive, spatial, and simulation-based methods. The intersectionality and climate vulnerability/resilience frameworks will guide this assessment, ensuring that selected models and algorithms align with the structural and SES-related inequities revealed in the evidence. Our intersectional framework will examine how gender, SES, and location share educational vulnerability (Baird et al., 2021), combined with the Intergovernmental Panel on Climate Change risk frameworks, and the Global Alliance for Disaster Risk Reduction and Resilience in the Education Sector Comprehensive School Safety framework to analyze policy responses and resilience strategies within educational systems (IPCC, 2022; GADRRRES, 2017). Concurrently, we will identify two focal provinces for deeper analysis based on data availability and quality, exposure to diverse climate events, socioeconomic diversity, and strategic policy relevance. This selection will be validated through early stakeholder consultations.

Stage 2: Climate Mapping and Evidence Review: Building on Stage 1 findings, the team will integrate temporal and spatial mapping of climate events and educational disruptions with an iterative review of interventions and policies documented in Pakistan and comparable LMIC contexts. This process will highlight trends, gaps, and lessons that can inform equitable educational planning and resilience strategies. Evidence from regional and international sources will provide benchmarks to situate Pakistan's responses in a broader context.

Stage 3: Initial Analysis and Dissemination: The team will conduct initial analyses of selected provincial datasets to explore how educational outcomes vary across SES groups and climate events, guided by an intersectionality lens and climate vulnerability/resilience frameworks (Ahmed & Eklund, 2024; Djoudi et al., 2016). These frameworks will structure identification of inequities, guide interpretation of administrative/policy data, and inform recommendations. Please see the figure below summarizing the overall methodology. Throughout all stages, findings will be validated through iterative stakeholder consultations to ensure policy relevance. The synthesis will explicitly document evidence strengths, limitations, and gaps to inform future research and climate-resilient education policy. This systematic approach will provide decision-makers with clear patterns of educational disruption and context-sensitive recommendations for ensuring equitable access during climate emergencies.



4.4. Proposed Timeline and Workplan (max. 100 words):

We will follow a structured five-month timeline across three interconnected stages. Month 1 establishes the project foundation through inception activities. Months 2-3 involve comprehensive literature mapping, data extraction, and province selection validated through early stakeholder consultations. Month 4 focuses on comparative review of interventions, integration of findings, and mid-point reporting to DARE-RC. Month 5 concentrates on evidence synthesis, preliminary analysis of provincial datasets, and all dissemination activities including policy brief development and stakeholder roundtable. The strategic overlap in the activities (see Gantt chart) will enable iterative refinement through emerging findings from subsequent stages to ensure robust synthesis and policy-relevant outputs. We will ensure that the final research report (comprehensive report of the research study), PowerPoint presentation of the research findings, and policy brief are submitted by 7 September 2026. The anonymised and de-identified primary raw dataset will be delivered by 31 August 2026.

Major Tasks and Activities	Month 1 (April)	Month 2 (May)	Month 3 (June)	Month 4 (July)	Month 5 (August)
STAGE 1: Systematic Document and Data Review					
Inception report and refinement of project focus	■				
Identification of policies, datasets, and grey literature	■	■			
Initial stakeholder consultations	■	■			
Province selection and validation		■			
Comprehensive literature and document mapping		■	■		
Data extraction and quality assessment		■	■		
Assessment of analytical models and frameworks			■		
STAGE 2: Literature Review and Mapping					
Temporal and spatial mapping of climate events			■	■	
Mapping of educational disruptions			■	■	
Comparative review of interventions (Pakistan and LMIC)				■	
Integration of findings with Stage 1 data assessment				■	
Mid-point report preparation and submission				■	
STAGE 3: Preliminary Analysis					
Preliminary analysis of provincial datasets				■	■
Stakeholder validation consultations					■
SYNTHESIS AND DISSEMINATION					
Evidence synthesis and drafting of scoping paper				■	■
Development of policy brief					■
Half-day stakeholder roundtable					■
Blog post(s) summarizing key findings					■
Finalization and submission of scoping paper					■

4.5. Risk Mitigation Plan (max. 100 words):

Risk	Likelihood	Impact	Mitigation Strategy
Limited access to administrative datasets or incomplete data	Medium	High	Treat accessibility and completeness as a key finding; use document review and grey literature to supplement gaps; engage provincial and federal stakeholders early to clarify availability. Clearly report gaps to inform future research and policy decisions.
Delays in policy or stakeholder engagement	Low	Medium	Schedule early communications with relevant authorities; provide clear briefings and context for data requests; build flexibility into Stage 1 timeline.
Ethical concerns in using sensitive education or climate data	Low	High	Only use anonymized, publicly available or officially shared datasets; adhere to data protection regulations; obtain permissions as needed.
Misinterpretation of findings by stakeholders	Medium	Medium	Present findings with clear caveats regarding limitations of datasets and evidence; provide concise, actionable policy briefs; conduct focused roundtable to clarify interpretation.

4.6. References:

- Ahmed, K. J., Haq, S. M. A., & Bartiaux, F. (2019). The nexus between extreme weather events, sexual violence, and early marriage: A study of vulnerable populations in Bangladesh. *Population and Environment*, 40(3), 303-324.
- Ahmed, S., & Eklund, E. (2024). Intersectionality shapes the access to various adaptive resources in climate-vulnerable contexts. *Environmental Research: Climate*, 3(4).
<https://doi.org/10.1088/2752-5295/ad8d04>
- Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19-32.
- Baird, S., Camfield, L., Ghimire, A., Abu Hamad, B., Jones, N., Pincock, K., & Woldehanna, T. (2021). Intersectionality as a framework for understanding adolescent vulnerabilities in low and middle income countries: Expanding our commitment to leave no one behind. *The European Journal of Development Research*, 33(5), 1143-1162.
- Djouidi, H., Locatelli, B., Vaast, C., Asher, K., Brockhaus, M., & Basnett Sijapati, B. (2016). Beyond dichotomies: Gender and intersecting inequalities in climate change studies. *Ambio*, 45, 248-262. <https://doi.org/10.1007/s13280-016-0825-2>
- Global Alliance for Disaster Risk Reduction and Resilience in the Education Sector (GADRRRES). (2017). *Comprehensive school safety framework 2017-2021*
<https://www.undrr.org/publication/comprehensive-school-safety-framework-2017>
- Intergovernmental Panel on Climate Change (IPCC). (2022). *Climate change 2022: Impacts, adaptation and vulnerability*. Cambridge University Press.
<https://www.ipcc.ch/report/ar6/wg2/>
- Peters, M. D. J., Godfrey, C. M., Khalil, H., McInerney, P., Parker, D., & Soares, C. B. (2020). Guidance for conducting systematic scoping reviews. *International Journal of Evidence-Based Healthcare*, 13(3), 141-146.
- United Nations Children's Fund. (2022). *Pakistan (Punjab) Education Fact Sheets I 2022: Analyses for learning and equity using MICS data*. UNICEF. https://data.unicef.org/wp-content/uploads/2022/12/2022Pakistan_Factsheet_2022p11.28.22-1.pdf
- UNICEF. (2023). *The climate-changed child: A children's climate risk index supplement*. UNICEF
- UNICEF. (2024). *Learning interrupted: Global snapshot of climate-related school disruptions in 2024*. UNICEF
- United Nations Children's Fund. (2023). *Is an End to Child Marriage Within Reach? Latest Trends and Future Prospects-2023 Update*. UNICEF
- World Bank. (2024). *Choosing our future: Education for climate action*. World Bank.

Documents to be Attached

5.1. CVs:

CVs attached.

5.2. Budget Template:

Budget using Excel template submitted.

5.3. Budget Justification:

One-page budget justification document enclosed.

5.4. Statements of Purpose:

Letter of Support attached.

5.5. Conflict of Interest Form:

COIs forms

5.6. Checklist:

Please check if the following supporting documents have been duly attached with your technical and financial proposals.

REMINDER to Applicant Organisation:

In addition to the documents listed above i.e. the CVs of core research team including Principal Investigator and Co-Investigators, Conflict of Interest Forms of the Principal Investigator and Co-Investigators, and the Budget and Budget Justification, the applicant organisation must also submit the following Supporting Documents, in accordance with Section 7 of the [Call for Applications Cycle-4](#).

1. Basic profile of the organisation focusing on completed and ongoing relevant research projects and other activities including the details like funding institution, research study area, geographical area, total budget etc. This is to be no longer than five pages.
2. Copy of Valid Registration Certificate (for private organisations) or a copy of Byelaws or relevant law, act or other document under which a public sector organisation is registered (for public sector organisations).
3. Complete auditor's report (including audited financial statements) for the years 2023 and 2024 (applicable to both public and private sector organisations).
4. In cases where public sector organisations do not have the above-mentioned audited financial statements and auditor's reports, they may submit any two audit reports issued by the relevant office of the Auditor General of Pakistan/or the relevant authority between 2022 and 2025.
5. Tax registration certificate.
6. Links to/submission of at least two recent peer-reviewed publications relevant to the priority areas and research topics given above: these publications can be authored by any of the key members of the research team.
7. CVs (not more than two-pages each) of core research team (including Lead Investigator/Researcher and co-investigators) - and list of publications of the Lead

Investigator/Researcher and co-investigators demonstrating experience of leading and conducting high-quality research in education/social sciences.

8. A confirmation on the applicant organisation's letterhead, signed by the competent authority of the applicant organisation, stating that it will obtain a No Objection Certificate (NOC) for conducting field data collection under the DARE-RC grant.
9. A confirmation on applicant organisation's letterhead, signed by the competent authority of the applicant organisation stating, "that the applicant organisation is not bankrupt or currently involved in any litigation related to bankruptcy".
10. A confirmation on applicant organisation's letterhead, signed by the competent authority of the applicant organisation stating, "that the applicant organisation is not suspended, or debarred, or otherwise identified as ineligible by FCDO, OPM, or any other organisation".

For details on the above supporting documents and their preparation and compliance, please refer to Section 7 of the [Call for Applications – Cycle 4 document](#).